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Review article

A critical review of the issue of cigarette butt pollution in coastal environments

Maria Christina B. Araújo[a,*], Monica F. Costa[b]

[a] Laboratory of Coastal Oceanography. Departamento de Oceanografia e Limnologia, Universidade Federal do Rio Grande do Norte, Praia de Mãe Luiza, S/N – Via Costeira, Natal, RN CEP: 59.014-100, Brazil

[b] Laboratory of Ecology and Management of Estuarine and Coastal Ecosystems. Departamento de Oceanografia, Universidade Federal de Pernambuco, Av. Arquitetura, Recife, PE CEP 50740-550, Brazil

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ABSTRACT

Beach anthropogenic litter is a worldwide problem and has been discussed in the specialized literature for decades. Cigarette butts (CB) are the most frequent form of personal item found on beaches. Yearly, 6 trillion cigarettes are smoked worldwide, and 4.5 trillion cigarettes are littered in the environment. The objective of our review was to assess the relevant literature on the issue of CB in costal environments, including urban areas. We compile and discuss studies (1998–2018) of CB sources for coastal environments, composition/degradability, quantification on beaches, toxicity to aquatic organisms and existing strategies to abate the problem. The literature shows that despite the growing interest in marine litter, this specific issue remains little studied and information is limited in time and space. Studies have been undertaken on islands, continental coasts, estuaries and coastal cities. There area wide variety of approaches to classification; for example, CB are considered plastic in 19% of studies and placed in an isolated category in another 16%. It was possible to identify the main sources of CB in coastal environments and access to the marine biota. In conclusion, we list and discuss proposals for reducing smoking, littering and marine pollution as a contribution to reduce the problems caused by CB in coastal and marine environments.

Capsule: Cigarette butts are a pervasive, toxic and recalcitrant type of marine litter that requires urgent attention from manufacturers, users, authorities and the public to prevent the ingestion of cigarette butts by biota and water pollution from its leachate.

1. Introduction

Beaches all over the world are full of anthropogenic litter. This problem affects inhabited areas (populated and unpopulated) as well as remote and isolated areas (Ivar do Sul and Costa, 2007; Ivar do Sul et al., 2011b; Becherucci et al., 2017; Cauwenberghe et al., 2013). Although marine litter is composed of a wide variety of waste types and there is a prevalence of plastic items (Scisciolo et al., 2016; Hengstmann et al., 2017; Suciu et al., 2017; Pasternak et al., 2017), cigarettes butts (CB), an item of mixed synthetic composition, stand out due to their global occurrence and significant amounts. According to Santos et al. (2017), 5505 trillion units of cigarettes were consumed in the world in 2016. By 2025, nine trillion cigarettes will be consumed worldwide.

Cigarettes butts (CB) are considered the most common form of personal litter in the world (Curtis et al., 2017; Novotny and Slaughter, 2014; Moerman and Potts, 2011). These residues are small, but are generated in large amounts. The CB problem has been reported for several years (Mindell, 2001; Santos et al., 2005; Oigman-Pszczcol and Creed, 2007; Martinez-Ribes et al., 2007; Bravo et al., 2009). According to the Surfrider Foundation (https://www.surfrider.org/programs/beach-cleanups), CB are the most frequently collected item during beach clean-ups promoted by the group. According to a report by the Ocean Conservancy (2011), approximately 52 million cigarettes were removed from coastal environments over 27 years. In the 2015 campaign, volunteers worked on beaches in more than 100 countries, collecting a total of 13,806,887 litter items, of which 2,127,565 were CB (Ocean Conservancy, 2016). Thus, this type of litter was at the top of the list of the most abundant litter items.

The goal of the present study was to carry out a survey of studies addressing CB involving questions related to their composition, quantification at beaches and coastal urban centres, main sources, impacts generated, and strategies used to control the problem.

* Corresponding author.
E-mail address: mcbaraujo@yahoo.com.br (M.C.B. Araújo).

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The study was intended to contribute to deepening the knowledge on the subject and to the diffusion of information about the different aspects of the pollution caused by this type of litter. The results obtained in this study will certainly allow the development of actions to mitigate the problem.

2. Results

2.1. Chemical composition

CB contain four main components: a filter; burned and unburned tobacco; ashes; and paper. Filters were added to cigarettes in the 1950s after increasingly convincing scientific evidence that cigarette caused lung cancer and other serious diseases, such as pulmonary emphysema. Over the past 50 years, almost all smokers (99%) started smoking filtered cigarettes (Novotny et al., 2009).

Cellulose acetate (the filter material) is a synthetic polymer made from cellulose (a natural polymer of vegetable origin) through a process known as acetylation that includes the addition of acetic anhydride and acetic acid. At the end of the process, plasticizers are added (such as polyethylene glycol). Cellulose is readily biodegraded by organisms that use the cellulase enzyme. Due to the modification of the polymer by chemical processes, the product obtained (cellulose acetate) is photodegradable, but has a limited potential for biodegradation. Moreover, the disintegration of conventional cigarettes is hampered by the high fibre compaction and addition of plasticizers (Puls et al., 2011).

There are over 5000 compounds present in cigarettes. Among these, at least 150 (of which 44 are found in large amounts) are considered to be highly toxic, mainly because of their carcinogenic and mutagenic potential (Hoffmann and Hoffmann, 1998; Slaughter et al., 2011). When burned, many of the chemicals present in cigarettes produce new compounds (Moriwaki et al., 2009; Novotny et al., 2009). The compounds with the highest toxic potential are mainly concentrated in the remains of tobacco and in the filter (Iskander, 1986; Slaughter et al., 2011; Barnes, 2011). These compounds can contaminate the soil after leaching by rainwater and are superficially transported to aquatic environments where they can be detected (Slaughter et al., 2011).

2.2. Amount collected

Assessments of the marine litter pollution in the world's coastal environments, focusing on several aspects of the problem, have systematically increased over the last few decades, with hundreds of published studies. However, surveys including the quantification of specific data related to CB are still very scarce and concentrated in a few regions (Fig. 1, Table 1; Electronic Supplementary material).

In the studies assessed, CB were classified into several categories (Fig. 2, Table 1; Electronic Supplementary material). Classification, as an isolated category, probably occurred because of its very specific composition, which mixes natural and synthetic components.

2.3. Sources of CB for beaches and coastal environments

The presence of CB on beaches is determined by a number of factors, including solar intensity, winds, currents, rivers and the frequency of beachgoers. Smokers' poor environmental behaviour (littering) and efficiency of cleaning services are other important determinants.

Environmental factors contribute to the transport of this type of litter since they are lightweight and can easily be transported by wind. The position of beaches in relation to the prevailing winds (leeward or windward) is very important in the accumulation of marine debris. Windward areas tend to accumulate more debris (Blickley et al., 2016; Scisciolo et al., 2016; Wilson and Verlis, 2017; Blickley et al., 2016). CB can float for long periods before becoming saturated with water and sinking, allowing their transport by rivers and currents (Engler, 2012), explaining why nicotine has been detected in aquatic environments such as lakes (Buerge et al., 2008) and rivers (Valcarcel et al., 2011).

Therefore, human influence is, direct or indirectly, also a strong determinant of CB accumulation on beaches. According to Novotny et al. (2009), the number of CB found on beaches is not necessarily related to the cigarettes smoked *in situ*. Transport to the sand from the pavement, roads and drains (Fig. 3) can also occur along a range of different distances.

Urban areas with high population densities have an important amount of CB litter in public spaces such as roads and pavements, especially around commercial areas (See "coastal urban centres", Table 1). In these areas, there are a larger number of selling points and smokers. The amount of CB litter near the sale and consumption areas is high, showing a strong correlation with product availability and consumption by locals (Moriwaki et al., 2009; Pon and Becherucci, 2012; Becherucci and Pon, 2014; Marah and Novotny, 2011). These areas may be a source of litter, including CB, for beaches nearby (Pon and Becherucci, 2012). If they are not adequately removed by cleaning services, they will be transported by wind and urban runoff through drainage systems from where they can reach coastal environments (Armitage and Rooseboom, 2000; Williams and Simmons, 1999).

Smokers are frequently reported as discarding their CB improperly (Rath et al., 2012; Novotny and Slaughter, 2014; Patel et al., 2013; Schultz et al., 2011; Smith and Novotny, 2011), but often do not recognize their littering behaviour as a pollution source, and justify it as a natural extension/continuation of smoking, regarding it as acceptable, and argue that CB are less important than other items, such as bottles, for instance. Their argument is that CB do not significantly harm the environment since they have the perception that they are biodegradable. Some feel uncomfortable littering, but see no alternative (Smith andNovotny, 2011). Non-smokers believe that CB are toxic more often than smokers (Rath et al., 2012)

Littering on the street can be a source of CB for beaches and coastal environments, but smoking on the beach is also a significant source that results in the large amounts found in the sand, where littering is very common. The intense use, especially during high season, is directly correlated to an increase in beach litter and CB (Santos et al., 2005; Ariza et al., 2008; Bravo et al., 2009; Hoellein et al., 2015; Silva et al., 2015; Simeonova et al., 2017; Becherucci et al., 2017).

Distance to urban centres is also a factor influencing the accumulation of beach litter, including the predominance of CB (Maziane et al., 2018; Leite et al., 2014).

Public cleaning services are also important in determining the levels of pollution of beaches by litter, and poor practices may favour their permanence and prevalence on beaches (Ariza et al., 2008; Ariza and Leatherman, 2012). Regular cleaning (mechanized or manual) by municipal services and episodic events arranged by groups of volunteers are usually inefficient in removing CB due to their small size.

In summary, many factors contribute to presence of CB on beaches, including natural aspects of the environment and also the behaviour of smokers in public places, both in the commercial and leisure centres in cities, and on the beach itself.

2.4. Impacts of CB

The harmful effect of cigarette smoking in both active and passive smokers is well known and is a public health issue. Cigarette smoking kills 6 million people each year, with 8 million deaths expected for 2030, resulting in over 1 billion people dead by smoking in this century (Mathers and Loncar, 2006). Natural environments and their biota are also at risk from the different phases of tobacco planting, processing, consumption and littering.

After the rapid deterioration of the labile fraction of the external paper wrapping, degradation proceeds very slowly, independent of environmental conditions (Bonanomi et al., 2015). Overall, studies

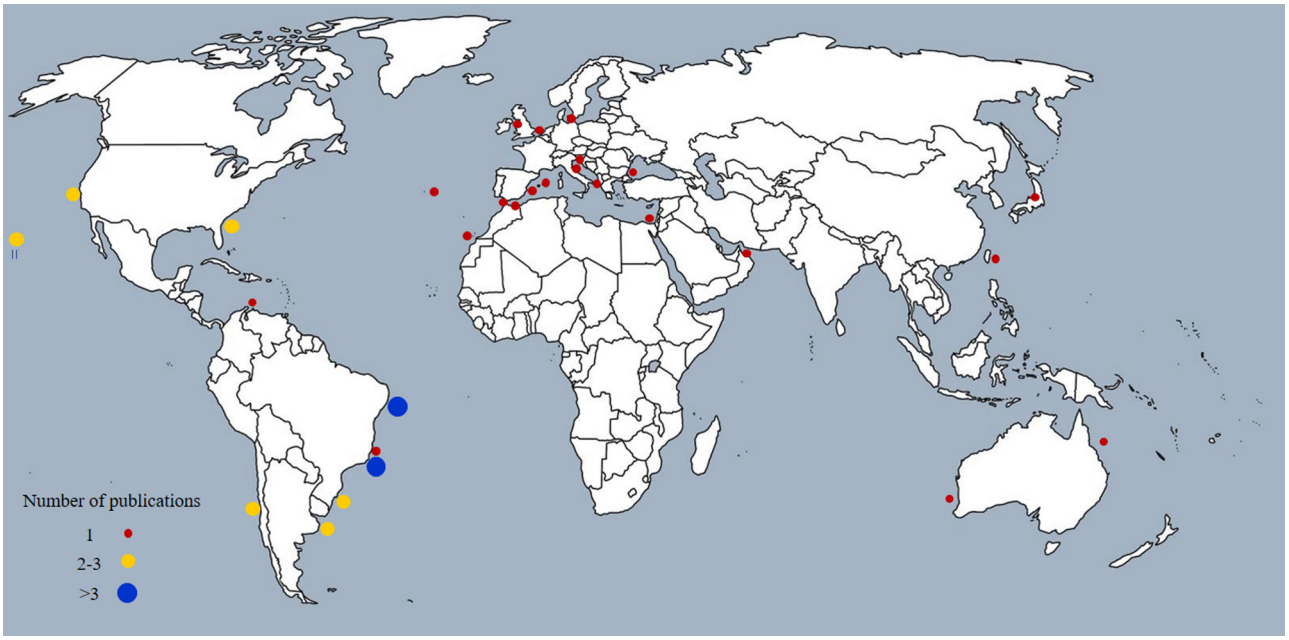


Fig. 1. Distribution of studies involving quantification of CB on beaches and coastal urban centres (1998–2018).

suggest that the longer CB stay in the environment, the greater the pollution caused. According to Green et al. (2014), one single CB can contaminate 1000 L of water. It is a relevant threat to the quality of urban waters, including supply reservoirs.

Two processes are mainly responsible for CB impacts in natural environments: the leaching of CB compounds by rainwater and its transference to water bodies through urban runoff. The main compounds include nicotine, aromatic polycyclic hydrocarbons and metals (Table 2).

2.4.1. Impacts on coastal environments

When accumulated on beaches, CB from direct littering or transported from other areas can cause different impacts on society, economy and environment (Fig. 4). In relation to leaching of metals from CB, it is studied mainly in freshwater systems, but there is a lack of information for coastal and marine environment (Dobaradaran et al., 2017).

2.4.2. Impacts on the biota

CB have been found in the stomach contents of marine fauna (fish, birds, whales) that accidentally ingested them during feeding (Santos et al., 2005a), as reported for *Chelonia mydas* and *Eretmochelys imbricata* (marine turtles) along the Brazilian Northeast coast (Macedo et al., 2011).

In addition to ingestion, chemicals present in cigarettes can also be harmful to aquatic organisms (Wright et al., 2015; Booth et al., 2015; Slaughter et al., 2011; Lee and Lee, 2015; Micevska et al., 2006; Savino and Tanabe, 1989).

The effects of nicotine on fish have been investigated for almost 50 years (Konar, 1970). The focus of the work was to determine if nicotine was a good anaesthetic for fish in fisheries, in an attempt to replace sodium cyanide and rotenone, which are potent poisons that were commonly used at that time. From a 40% nicotine solution, dilutions were tested on fish and were association with diverse symptoms such as acute poisoning, palsy of gills, convulsion and death.

More recently, studies remain limited in respect to number and test-organisms used (Table 3), but the focus of the works is to assess the effects of substances in the tobacco and in cigarette filters on the environment and the organism, both directly and indirectly.

In summary, tobacco contains thousands of potentially harmful compounds, and CB, when exposed in the environment, liberates these substances, including nicotine, trace metals and numerous other compounds. After smoking, filters can damage the environment by acting as vectors for toxic chemicals such as trace metals, nicotine and carcinogens to aquatic habitats; also, they can be ingested and release these substances within the animal's body. In addition to the impacts on the environment, CB pollution is also associated with serious social and economic damages.

2.5. Strategies for reducing CB in the environment

2.5.1. Smoking ban in public places

Strategies for reducing CB in the environment often include the ban of smoking in public spaces, including beaches, where second-hand smoking, aesthetics and environmental consequences of cigarette toxicity are major concerns (Novotny et al., 2009; Johns et al., 2015). Banning smoking in public open spaces results in significant reduction of cigarette-related litter (Axelsson and van Sebille, 2017; Johns et al., 2011, 2013; Bayer and Bachynski, 2013; Ueda et al., 2011) and transport of CB to aquatic and coastal environments is reduced (Healton et al., 2011). Resistance is met from politicians and related groups who claim banning smoking in public places is a violation of individual rights and no more a source of CB on beaches than urban runoff. In addition, compliance depends on resources that are not always available, especially in countries where there are other investment priorities. Finally, smokers do not want to feel apart (Ariza and Leatherman, 2012).

One of the earliest smoking bans on beaches was at Hanauma Bay (Hawaii) in 1993. Both aesthetics and environmental arguments motivated community and park staff to embrace this measure. Its on-going success encouraged the adoption of similar restrictions at other sites (Ariza and Leatherman, 2012). Since then, laws banning smoking in public spaces such as beaches and parks have been widely adopted in the United States, especially in the last decade. New York joined other hundreds of US cities in 2011 and banned smoking at all its parks and beaches. A significant decrease in the number of smokers at parks and beaches was then detected by the public perception (Johns et al., 2015). Presently, according to the American Non-smokers' Rights Foundation (ANR; https://no-smoke.org/), all beaches of 317 municipalities in the United States are smoking-free by law.

Table 1
Literature review focused on the presence of CB in coastal environments where it can cause environmental and toxicological damage to marine biota. See also Electronic Supplementary Material for more detail on these references.

Environment	Reference	Location	Total litter (number of items)	Classification of CB	Most abundant items (%)
ISLANDS	Scisciolo et al., 2016	Aruba (Caribbean)	42,585	Plastic	Plastic 89.24%, CB 6.78%
	Smith et al., 2014	Rottnest Island (Australia)	4,520 (40 volunteers)	Isolated category	Plastic 26.2%; Glass fragments 9.5%; CB 5% and fifth most abundant item
	Wilson andVerlis, 2017	Islands of Wreck, Tryon, Heron and Northwest (Australia)	130 to706 items/month (28 months)	Plastic	Plastic 69 to 95% CB up to15% (on most visited island)
	Baztan et al., 2014	Lanzarote, La Graciosa e Fuerteventura (Canary Islands)	Focus on microplastic. 1 g/l to40 g/l of sediment	Not classified	CB most abundant item of macro-litter on beaches (does not report %)
	Hengstmann et al., 2017	Rügen Island(Germany)	1,115	Paper (according to OSPAR, 2010)	Plastic 82.7% CB 15%
	Pieper et al., 2015	Faial Island (Azores, NE-Atlantic, Portugal)	28,261	Plastic	Plastics 93.14%, CB 2.09%
	Blickley et al., 2016	Maui Island(Hawaii)	3,941	Isolated category	Plastics 71% - 94% CB45%
	Carson et al., 2013	Hawai'i Island	Not informed	Plastic	Plastics 73.6%, CB 1,267 items (most abundant item)
	Martinez-Ribes et al., 2007	Mallorca, Menorca and Ibiza (Balearic Islands-Spain).	11,231	Not classified	Plastics 36%, CB 56%
	Prevenios et al., 2018	Corfu Island (N. Ionian Sea)	41,617	Plastic	Plastic > 90% CB among the 10 most abundant items (does not report %).
CONTINENTAL COASTS	Bravo *et al.*, 2009	Chilean Coast	21,146	Isolated category	Plastic 27.1% CB 2nd most abundant item (does not report %).
	Hidalgo-Ruz et al., 2018	Chilean Coast	Not informed	Isolated category	Plastic (does not report %) CB in 2012: 38%; in 2016: 41.8%
	Becherucci et al., 2017	Mar del Plata and Villa Gesell (Argentina)	9,620	Isolated category	Plastic 34.31%, CB 53.24%
	Moore et al., 2001	Orange County (California-USA)	106 million	Isolated category	CB 4th most abundant item, 139,000 units.
	Ribic, 1998	New Jersey (USA)	Ranged between 179 and 3,198 items/year	Plastic	Plastics~ 396 items/month, CB ~58 items/month
	Williams et al., 2016	Cadiz (Spain)	2,277	Plastic	Plastic 71.62%, CB 8%
	Arizaet *al.*, 2008	Costa Brava (Spain)	Not informed	Not classified	CB 127,805 units
	Nelms et al., 2017	British Coast	2,376,541	Isolated category	Plastic Fragments (> 2.5 cm) 13% Plastic Fragments (< 2.5 cm) 10% CB 3% (among 20th most abundant)
	Cauwenberghe et al., 2013	Belgian Coast	51,428	Plastic	Plastic 95.5%, CB 0.5 to0.8%
	Simeonova et al., 2017	Bulgarian Coast	19,790	Paper (according to OSPAR, 2010)	Synthetic polymers 84.3%, CB27%
	Kuo and Huang, 2014	New Taipei (Taiwan)	9,319	Plastic	Plastic 33.58%, CB 6.01%
	Maziane et al., 2018	Mediterranean coast of Morocco	8,021	Plastic	Plastic 66.77%, CB 26.34%
	Pasternak et al., 2017	Israel (Mediterranean Coast)	69,122	Plastic	Plastic 90%, CB 12%
	Munari et al., 2016	Adriatic Sea (Mediterranean Coast)	2,502	Plastic	Plastic 81.1%, CB 22.9%
	Laglbauer et al., 2014	Slovenian coast	5,870	Plastic	Plastic 64%, CB 44.8% (28.7% of the total)
	Claereboudt, 2004	Oman (Arabian Peninsula)	3,939	Organic	Plastic 61%, CB 9.2%
	Oigman-Pszczol and Creed, 2007	Rio de Janeiro (Brazil)	15,832	Paper	Plastic 36.8%, Paper > 50% (CB 87%)
	Suciu et al., 2017	Rio de Janeiro (Brazil)	20,040	Plastic	Plastic > 80%, CB 10.4%
	Silva et al., 2015	Niterói (Rio de Janeiro-Brazil)	27,372	Plastic	Plastic 52.71%, CB 17.89%
	Silva et al., 2016	Niterói (Rio de Janeiro-Brazil)	2,789	Plastic	Plastic 89.77%, CB 5.4%
	Tourinho and Fillmann, 2011	Rio Grande (Rio Grande do Sul-Brazil)	29,717	Isolated category	Plastic 47.9%, CB 13.6%
	Santos et al., 2005	Rio Grande (Rio Grande do Sul-Brazil)	248,400	Isolated category	Plastics 23–25%, CB ~ 40%

(*continued on next page*)

Table 1 (*continued*)

Environment	Reference	Location	Total litter (number of items)	Classification of CB	Most abundant items (%)
	Portz *et al.*, 2011	Xangri-Lá (Rio Grande do Sul-Brazil)	1,390	Isolated category	Plastic 42%, CB 39%
	Silva-Cavalcanti et al., 2013	Recife (Pernambuco-Brazil)	165,882	Plastic	Plastic 74.3%, CB 19.78%
	Dias Filho et al., 2011	Recife (Pernambuco-Brazil)	20,090	Paper	Manufactured Items 57.90% (Plastic 43.3% / CB 22%)
	Leite et al., 2014	Salvador and surroundings (Bahia-Brazil)	7,858	Plastic	Plastic 87%, CB 8%
	Ivar do Sul et al., 2011a	Bahia (Brazil)	6,751	Isolated category	Plastic 52 to77%, CB 19.3%
	Andrades et al., 2016	Espírito Santo (Brazil)	4,752	Isolated category	CB most abundant item < 5 cm (17.6%) on the most urbanized beach
ESTUARIES	Viehman et al., 2011	Salt marshes (North Carolina)	14,747	Plastic	Plastic 45.6%, CB 5.6% (within 10th most abundant items)
COASTAL URBAN CENTERS	Pon and Becherucci, 2012	Mar del Plata (Argentina)	20,336	Isolated category	Plastic 22%, Paper 31%, CB 33%
	Becherucci and Pon, 2014	Mar del Plata (Argentina)	13,503	Isolated category	Plastic 19.7%, Paper 29.8%, CB 42.8%
	Marah and Novotny 2011	San Diego (California-USA)	Not informed	Isolated category	Mean number of CB collected was 38.1 (for locations with sale and/or consumption; n = 25) and 4.8 (for locations without sale and/or consumption; n = 26)
	Moriwaki et al.,2009	Ueda (Japan)	~ 690 items/month	Isolated category	Plastic ~ 98 items/month, CB ~ 490 items/month

Where smoking is forbidden, there are fines for those who disrespect regulations (Axelsson and van Sebille, 2017). Fines are usually effective (dissuasive) in reducing CB littering in public places (Axelsson and van Sebille, 2017; Ueda et al., 2011; Novotny et al., 2009), despite being criticized by smokers as a violation of their individual rights (Ariza and Leatherman, 2012).

A suggestion to return to non-filtered cigarettes was made by some tobacco company employees, but it was not accepted by the industry since it was considered more viable to invest in biodegradable filters than convincing smokers that non-filtered cigarettes were safe enough to smoke, or for the environment (Smith and Novotny, 2011). Hence, CORESTA (Tobacco Industry's International Research Organization) proposed the use of biodegradable filters as an alternative to the environmental problem, understanding that it could reduce disapproval by smokers and an improvement in the manufacturer's public perception (Novotny et al., 2009; Smith and Novotny, 2011).

Banning cigarette filters for environmental reasons could make sense; however, they continue to be used as a marketing asset, a suggestion that risks are therefore reduced (Novotny and Slaughter, 2014; Wallbank et al., 2017). However, contrary to common belief (especially among smokers), cigarette filters do not protect smokers' health (Healton et al., 2011).

To be effective and lasting, smoking bans must be understood and accepted by society as a whole, which requires educational campaigns based on scientific facts (Ariza and Leatherman, 2012).

2.5.2. Public policies

The *Framework Convention on Tobacco Control*[1] is the first treaty developed under WHO rules and offers member states a legal framework and guidelines to orient anti-tobacco policies, among them MPOWER (www.who.int/tobacco/mpower/en/), a group of measures of control, monitoring, health information for users and commercial regulation of tobacco products (Levy et al., 2016; Jackson-Morris and Latif, 2017).

WHO recommends the Parties to observe guidelines from its experts working groups that coalesce knowledge based on sound scientific literature in order to abate and improve all issues generated by the tobacco life cycle. For instance, Articles 17 and 18 of its main most recent directives suggest sellers should be offered technically and financially viable alternatives to deal with their wastes; thus extending the responsibility for tobacco products and wastes control to the environment as much as to people (WHO, 2003). This would be a simple enough task, if only the selling-consumption-littering triad was not so spread in space and time across all environments. So, it depends on governments and private managers to offer adequate disposal options, but to people of all backgrounds to use them.

To date, 179 countries and the EU accept MPOWER; Brazil, Thailand and Turkey are highlighted for implementing all, or almost all, lines of actions proposed in the documentation (Levy et al., 2016). Turci et al. (2017) reports that the greatest difficulty in effective implementation of anti-tobacco measures based on the WHO Framework Convention on Tobacco Control is the strong manufacturers and traders lobby.

The 'Index of Tobacco Control Sustainability' (ITCS) developed by Jackson-Morris and Latif (2017) is a tool to assess if countries have policies, structures, resources and capacities to effectively support anti-tobacco measures (2017). Tested in countries representative of each WHO region (China, Chad, Indonesia, Mexico, Bangladesh, Georgia and Pakistan), it has 31 indicators and was capable of rising the sustainability issue. Participants from different groups of stakeholders were probed in their local languages, and the majority considered all 31 indicators "critical" and "important". "Critical" indicators were related to funding national measures, social responsibility of the sector's

[1] Adopted May 21st, 2003, and enforced since February 27th, 2005.

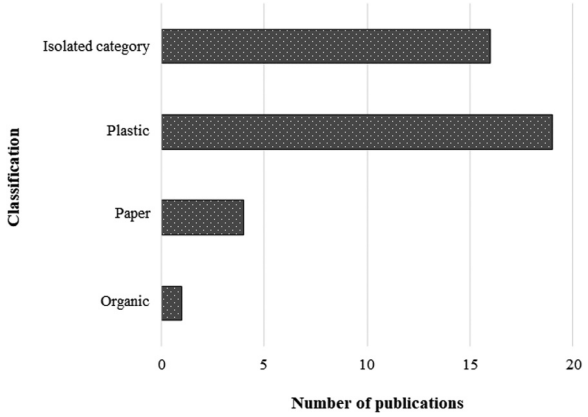


Fig. 2. Categories of classification of cigarette butts in the studies assessed (1998–2018).

industry and acquisition of national-level data on mortality and morbidity, among others.

2.5.3. Taxes and fees

Since the price of cigarettes does not include the costs to the public sector of manual and mechanical cleaning of public spaces and disposal of its wastes, or the costs related to its damage to biota and ecosystems, the pollution caused by CB littering is a negative externality. Estimates of the total costs (annual) to cities of litter abatement range from US$3 million (city with 1 million inhabitants) to US$16 million, in Toronto (Schneider et al., 2011). These estimates do not include costs to public health systems (Schneider et al., 2011). In this way, the whole population pays to remediate the consequences of smoking and littering through their taxes.

The addition of a fee or a tax to the price of cigarettes has been adopted in some places as an alternative to finance costs related to smoking. In 2009, a US$0.20 fee was added to the price of cigarettes in San Francisco. In January 2010, Philip Morris sued the city requiring its extinction (Smith and McDaniel, 2011; Smith and Novotny, 2011). The tobacco industry consistently defends that the responsibility to correctly discard cigarette residues is on the smoker, and claims that educational and campaigns and behavioural changes would suffice, with product taxation not being necessary. The industry implicitly defends that pollution is a result of smoker's behaviour and that they have no direct influence on it. Anti-littering campaigns supported by the industry did not change smokers' poor habits, but the sector insists on education and the availability of ashtrays (both public and individual models) (Smith and McDaniel, 2011).

2.5.4. Recycling

There is no known financial value in the recycling of CB (Barnes, 2011). However, due to the rising awareness of the environmental problems caused by this type of litter, a method to extract cellulose and then recycle paper from CB collected in natural environments was developed (Teixeira et al., 2017). It was possible to prepare different paper sheets, using cellulose acetate without previous treatment and CB cooked in an alkaline medium. The authors suggest that the resulting product could be interesting to the recycled paper industry pure or mixed with ordinary paper pulp in different amounts, depending on the requirements of the final product. Therefore, the results point towards a friendly alternative to this sort of waste to avoid littering, as well as resolving part of the environmental problem caused by this pollutant.

In São Paulo city (Brazil), where an estimated average of 34 M CB are generated each day, a consortium of companies (*Bituca Verde*, or Green Butts and *Renova Ambiental*, or Renew Environment) began a business using littered CB. The material is burned as fuel in furnaces of cement and steel plants. The paper is also used by the recycled paper industry (www.bitucaverde.com.br; in Portuguese).

2.5.5. Handing out portable ashtrays

Handing out individual portable ashtrays, especially at beaches, has been advocated for a number of occasions as an alternative where there is no infrastructure to receive this (or any other) type of waste and smoking is allowed. These items are usually made of hard recycled plastic and are reusable many times over (Widmer and Reis, 2010).

At Lanarca (Chipre), visitors can help themselves to plastic cones available at the beach entrance for their convenience during their stay at the beach (Figs. 5a and 5b). Upon leaving, the cones must be emptied in a larger container and left on the support for the next visitor. The effort also aims at improving environmental awareness, cleanliness and hygiene of the beaches. A similar initiative was also implanted at Praia do Rosa (Brazil) (Fig. 5c). This sort of individual container exists in many different shapes, sizes and materials (Fig. 5d).

This practice was tested (Widmer and Reis, 2010) using the cone model at a cost of US$1/day. The authors made a stratified sample scheme that considered different days of the week during a fortnight. The results showed that the cones were adequately used for CB and other small items that would not be taken by the beach cleaning



Fig. 3. Urban areas as possible sources of cigarette butts to beaches and coastal environments.

Table 2
Brief review of CB chemical contamination of natural environments.

Compounds present		Reference
Nicotine	Madrid (Spain), river waters are contaminated by nicotine with up to 1.9×10^{-3} mg L^{-1}.	Valcarcel et al. (2011)
	In Switzerland, 10 water treatment plants and 8 lakes were proved to be contaminated, showing that nicotine can be transferred from one environment to another.	Buerge et al. (2008)
	CB littering was assessed in Berlin (Germany), as a source of nicotine to adjacent water bodies through urban runoff. In the lab, the effect of rainfall over CB was modelled and nicotine elution was observed since the start of the washing process and, after 15 events (1.4 mm of rainfall each) the nicotine concentration in the water was 3.8 mg g^{-1}, with the first event responsible for 47% of this amount.	Green et al. (2014)
Aromatic polycyclic hydrocarbons and metals (incorporated by plants from the soil)	An elution test of the CB (obtained from the roadside in Ueda city) was carried out. The content of heavy metals was more than the detection limits for all elements, cadmium, copper, lead, chromium and arsenic. Also, polycyclic aromatic hydrocarbons (PAHs) from cigarette butts into the environment were confirmed.	Moriwaki et al. (2009)
	The analysis of some metallic species (As, Pb, Cd, Cu, Ni, Cr, Co, Al, Mn, Zn, and Fe) showed that these species are essentially attached to the nano scale particles from CB by leaching with rainwater.	Chevalier et al. (2018)
	The amounts of Ba, Cd, Cr, Cu, Fe, Mn, Ni, Pb, Sr, Ti, and Zn in CB without tobacco residues was tested using water leachates during 24 h, when all elements were then detected.	Moerman and Potts (2011)
	Cd, Fe, As, Ni, Cu, Zn and Mn contents were found at nine stations along the northern Persian Gulf in coastal areas, showing that these metals can indeed enter the marine environment from CB leachates.	Dobaradaran et al. (2017)



Fig. 4. Some impacts of cigarette butts on beaches.

services. Only 1.5% of the cones were left on the beach.

Despite the goodwill of smokers who volunteered to use the cones (or other containers) within this model of responsible waste management, efficacy remains questionable since, on occasion, there were "30 CB inside the larger container and 100 around it" (Smith and Novotny, 2011).

2.5.6. Campaigns targeting beach users

The tobacco industry strategies include socioenvironmental actions as public relations tools to gain credibility and favour sales. Some of these actions include beach clean-ups during which brands are promoted globally. In Greece, they work with the Hellenic Marine Environment Protection Agency (HELMEPA) in public campaigns concerning smoking and environmental pollution. The alleged objective of the campaign is to safeguard the seas against CB pollution; in the meantime, they advertise their brands through delivery of materials and ashtrays with the logos of tobacco companies to thousands of people (Vardavas et al., 2010).

Eastman et al. (2013) assessed the behaviour, knowledge and drive of people to solve the problem of beach litter in Chile. Students (909) from 38 schools participated and concluded that the best possible strategy to reduce beach litter was education. Improvement in infrastructure (bins), fines and different/more intense cleaning schemes were also mentioned. The authors suggest that campaigns should have multiple targets for better results; for example, involving students in beach cleaning campaigns that associate education and the enhancement of public perception of the problem.

Many organizations broadcast information on the hazards of CB littering on beaches and coastal environments to call attention to smokers and non-smokers. Materials aim at beach users, in general, and call attention to the consequences of this type of solution, as an attempt to reduce littering (Fig. 6). In South Korea, for instance, the government intensifies campaigns during the summer to reach people during their holidays. Their materials try to value that particular time of the year, and present the use of healthy environments as a source of pleasure and a reward. Flyers read "*Are you going on holiday? Leave your cigarettes at home*" (Simpson, 2005).

At Bayfield, Grand Bend, Waubuno and Wasaga (Blue Flag beaches - Canada), an initiative of the *Pacific Whale Foundation*[2] called *Keep Your Butts Off the Beach* was started in 2001, and incentivizes smokers to responsibly manage their CB and distributes free individual containers.

2.5.7. Beach clean-ups

Beach clean-up is usually by hand, using a rake and large sieves for transport; sometimes small mechanized vehicles are used as a complement. It is necessary to remove all debris from the site by bagging and road transport.

Municipalities usually hire this service and where labour is still affordable, the manual collection model prevails (*e.g.,* Brazil), being normally intensified during summer and vacation months. Out of season, the service might be completely interrupted. The mechanized option uses a type of tractor with a railing tool that ploughs the dry sand areas. This option is more frequent where labour is more expensive. The size of the rake teeth and tractor rail mesh also determines the efficiency of the job.

Both systems are relatively inefficient for small items, especially CB, seals, bottle caps, and fragments (Widmer and Reis, 2010). Manual cleaning usually relies on the person's vision and physical condition,

[2] An NGO dedicated to the promotion of ocean literacy, with a special focus on whales, dolphins and coral reefs.

Table 3
Toxicology and main impacts of tobacco and cigarette butts leachates on aquatic biota.

Tested for	Species used	Main results	Reference
Effect of compounds present in CB (including nicotine) on the growth of crustacean.	*Daphnia pulex* (Cladocera)	The compounds significantly reduced growth	Savino and Tanabe (1989)
The acute toxicity of 19 types of cigarette (with filters) of six brands containing different concentrations of tar and nicotine.	*Ceriodaphnia dúbia* (crustacean Cladocera) *Vibrio fischeri* (marine bacterium)	General increase in toxicity (decreased EC50 values) for both test-organisms with increasing tar levels	Micevska et al. (2006)
Effect of nicotine exposure on mobility in planarian	*Dugesia dorotocephala*	Nicotine causes a number of adverse effects, including decreased motility.	Rawls et al. (2011)
Exposure to a solution made from CB extract to assess its genotoxicity on bacteria	*Salmonella typhimurium* *Escherichia coli*	The effect was highlighted in all the bacterial strains tested, suggesting the ability of toxicants contained to induce frameshift mutations, base substitutions, and oxidative damages.	Di Giacomo et al. (2015)
Lethal and sub-lethal response of molluscs Gastropoda to chemicals found in cigarettes (leachates produced from CB immersed in seawater with dilutions of 5%, 10%, 25% and 100%)	*Austrocochlea porcata,* *Nerita atramentosa* *Bembicium nanum*	All individuals died in the 100% solution. Alterations were observed in all species and concentrations tested. Different behaviours were observed at concentrations, including escaping, search and antenna movements.	Booth et al. (2015)
Effect of compounds present in CB filters and its fibres on the behaviour and physiology in marine worm.	*Hediste diversicolor* (Polychaeta from coastal sediments)	It was observed a significant inhibition of behaviour of the animal. Higher concentrations resulted in bigger DNA damage and reduced growth rates.	Wright et al. (2015)
If CB with and without tobacco and smoked and non-smoked differ in leachate toxicity; if filters and tobacco differ in toxicity and; if smoking increases the toxicity of the filters leachate.	*Atherinops affinis* (marine fish) *Pimephales promelas* (freshwater fish)	Results allowed establishing a decreasing order of toxicity: smoked butts with tobacco > smoked filters without tobacco > non smoked filters. Tobacco alone was more toxic than the naked filter.	Slaughter et al. (2011)
Effects of compounds present in smoked and non-smoked cigarette tobacco and filters on larval development	Larvae of *Oryzias latipes* (freshwater fish)	Low concentrations of the leachates raised the heart rate, accelerated development, and changed behaviour, while high concentrations lowered the heart rate, suppressed development, and increased mortality	Lee and Lee (2015)

and therefore concentrates on large highly visible items as plastic bags, cups, bottles and food containers. CB are small and light (sometimes sand)-coloured. The rail mesh size is calculated to operate for long distances and times without the need of cleaning, so it is too large for CB (~3 cm) and other debris of similar size. Therefore, these items remain in the sand, regardless of cleaning efforts, possibly being buried over time (Ariza et al., 2008; Ariza and Leatherman, 2012).

Therefore, the use of funds for beach cleaning raised by taxing tobacco products over wide scales of time and space remains questionable, despite its increase as a compensation option (Smith and McDaniel, 2011; Vardavas et al., 2010). Use of these funds can impair the search for more effective and definitive solutions when compared to the palliative and temporary effort of their simple removal.

Volunteer engagement in beach clean-ups started in 1985 and is growing at exponential rates since the turn of the XXIst century. The most well-known programs are *Clean Up the World* and *Annual International Coastal Clean up*. Both work on the "think global act local" logic.

Clean Up the World is run in consortium with the UNEP and aims at capacity building for environmental conservation at the local level. This program mobilizes approximately35M people in 130 countries every year. The *Annual International Coastal Clean up* brand belongs to *The Ocean Conservancy,*[3] and in the last 30 years has collected over 215 M kilograms of debris from beaches and other environments around the world. Both campaigns have a world register of results available for the public.

A number of isolated actions partly inspired by these two large initiatives happen around the world each year, and numbers of places and people and tons of debris are growing. The method is also used for wide region scientific assessments, as in the UK, where 736 beaches were studied based on such proxies (Nelms et al., 2017).

In summary, there are a number of proposals to face the problems caused by the presence of CB on beaches that include preventive and corrective actions, as well as different public and private stakeholders. All of them can be adopted together, depending on the availability of resources, infrastructure and adequate legislation.

3. Discussion

Recently, the focus of marine pollution issues has been directed to plastic marine debris. In 2009 alone, residents of the coastal zones of the world might have been responsible for the arrival of over 9 M tonnes of plastic debris to the oceans (Jambeck et al., 2015). Within this scenario, CB found on beaches have gained attention from users, NGOs, governments and researchers dedicated to studying marine pollution.

Usually, people are strict about the environmental quality of the open spaces they use, and litter on the sand and in the water is one of the top 5 parameters users consider when choosing where to go for leisure and relaxation (Vaz et al., 2009; Ariza and Leatherman, 2012; Botero et al., 2015). Beach tourism is often a strong component in the revenue of many communities and regions and therefore it is expected that litter can cause loss of aesthetics, safety and desire to use the space. Consequently, it will interfere with users' satisfaction and revenue, eventually leading to the abandonment of the beach as a leisure option.

CB on the beach is a well-known fact, omnipresent and easily detectable by users and researchers alike. An estimated 4.5 of the 6 trillion cigarettes consumed every year are discarded in the environment (Novotny and Slaughter, 2014). Very likely, there is an unaccountable amount of CB in beach sands all over the world, which are inaccessible to conventional cleaning services. However, source identification and characterization is still a fundamental question to solve the problem (Pasquini et al., 2016) and cannot be underestimated in plans and policies; otherwise, any cleaning attempt will be in vain.

The second half of the XXth century marks the popularization of filtered cigarettes, as a part of urbanization, changes in production models (work *vs*. capital) and consolidation of a consumer market economy (Boeira, 2002). Cities promote more intense public space interactions, and both public and private infrastructure to support people to stay at such places is required. Usually, the selling of cigarettes and tobacco-related products is concentrated at such places; consequently, littering of CB is also more intense around gathering places, which in

[3] An NGO based in Washington D.C. (USA) founded in 1972.



a



b

Source: http://cyprus-mail.com/2017/05/29/new-eco-ashtrays-butt-free-beaches/



c

Source: http://dc.clicrbs.com.br/sc/noticias/de-ponto-a-ponto/noticia/2015/12/iniciativa-recicla-bitucas-de-cigarro-na-praia-do-rosa-em-imbituba-4940952.html



d

Source: http://blog.kindred-spirit.net/2015/10/13/north-carolina-beach-towns-wrightsville-carolina-beach/

Fig. 5. Beach ashtrays models of use and variety.

coastal cities includes beaches. There is evidence that smokers who use beaches are also responsible for the accumulation of CB in sand, just as much as transport from other areas of the city. During recent history, smokers have associated putting out cigarettes with sand; therefore, the habit of littering on beaches is automatic.

The prevalence of this habit is also associated with other factors, such as poor law enforcement regarding littering, in general; absence of adequate penalties; and poor advertisement of the environmental problems caused by tobacco products, despite the focus of public campaigns on health issues for both smokers and second-hand smokers. A serious lack of vision of the entire life cycle of tobacco, its products and their resulting wastes makes anti-tobacco actions and pollution control difficult.

Additionally, the use of beaches for leisure and pleasure might cause an unconscious distancing feeling, a relaxation that might lead smokers to forget their citizen duties towards the commons and the global environment.

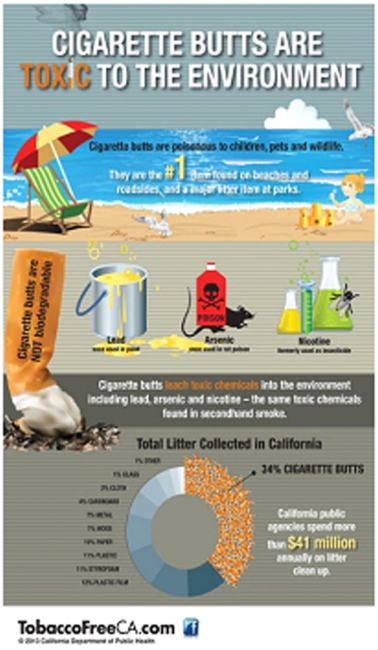
Despite tobacco industry resistance, the health risks to smokers and second- and third-hand smokers has been slowly recognized by the scientific literature and beyond (Boeira and Guivant, 2003). Presently, there is a global mobilization against tobacco production, manufacturing and use to decrease health risks. However, according to Booth et al. (2015), the environmental concerns do not follow with the same strength and speed.

The risks posed by CB to the aquatic biota remain underestimated because they are poorly known, both from the qualitative and quantitative aspects. Specific toxicity and physical damage by contact and ingestion are not well defined for aquatic fauna, let alone flora. Research into these topics is urgent if we are to build a case against a powerful economic player such as the tobacco industry. In the case of ingestion of debris, referring to marine vertebrates, CB offer a risk of total or partial digestive tract blockage, despite their malleability, contributing to transit and elimination issues. In addition, the CB can leach diverse toxic substances inside the body of the animal, leading to toxicity. It is interesting that ingestion can happen many times during the animal's life, and each event lasts for days inside the digestive tract before elimination, enough time to exchange substances with food contents and faeces (which ultimately are encountered by other animals).

Prevention, control and mitigation of impacts caused by CB in the marine environment has proven to be a difficult question that requires articulated strategies, public and private resources and much public participation. A number of proposals have been made and tested around the world, but so far they are relatively isolated, and depend on diverse



Source:
http://www.sdcoastkeeper.org/blog/environmental-education/cigarette-



Source:
http://tobaccofreeca.com/environment/cigarette-butts-are-



Source:
http://environmentaldefence.ca/2016/09/15/mind-not-sight-beach-litter-problem

Fig. 6. No-littering campaigns aimed at cigarette butts.

factors to work minimally. Stakeholder's support is a key issue and the role of the tobacco industry is no less important.

From the many proposals aiming at reducing smoking, risks to second-hand smokers and littering, the most controversial is banning smoking from public spaces, including beaches. Debate involving smoking enters the ethics of individual rights (Boeira and Guivant, 2003), leaving public health risks for the entire population and environment second. How far can we go accepting risks to guarantee individual rights? What is the responsibility of industry? When should risk perception by the public be replaced by scientific risk assessment? All these questions remain open and we urge that they be approached by society with maximum seriousness and speed. There is an environmental information gap to be filled and a scenario to be shown to society that could put smoking on a list of crimes against humanity and the future of biodiversity, yet another ethical issue.

The argument that they are responsible for what happens to their own bodies is no longer a hiding place for smokers. Evidence points towards a greater responsibility than that on social and environmental planes. Although a programmed complete ban on smoking could be seen as an abuse of individual rights, the costs inflicted on society and nature are per se even more abusive and irreparable. The cause suggests smokers would have to comply with the ban in favour of more prominent issues and positive externalities to themselves.

Ideally, this complete ban should start being enforced in regions where political, financial and social infrastructure could support the impact of job losses to philosophical questioning of individual rights; the need to revolutionize secular land use for strict monitoring of effects to coastal environments; and clinical support for the ecotoxicological evaluation of water and other natural compartments. Public support would have to be guaranteed through campaigns and educational actions that could inform about the pros and cons of this choice, should it be implemented. Meanwhile, the infrastructure must be adequate to cope with the delay in making this decision.

Environmental education is a slow process, although it is an important asset in changing behaviour towards the environment and society, and allows environmental quality to be perceived as a common daily goal.

On beaches where smoking is still permitted another question arises: the need for practical alternatives to litter. The individual container is a start, but it is not scalable to beaches that receive thousands (eventually millions) of people in a day. In addition, there is the issue of safety since CB can cause small fires when thrown into containers while still lit – therefore the idea that they should be put out in the sand.

Following a partially successful experience in Brazil, where pictures of smoking damage to health are bold and bluntly printed on the cigarette pack, information about environmental damage could also be part of this set of images and information. Providing information that indicates that CB pollute the environment, that filters ARE NOT biodegradable, that CB can cause real physical and toxicological harm to fauna and flora, and that pollution impairs tourism and leisure could start making smokers thinking that, perhaps, cigarettes are not such a good idea after all.

Although littering is a personal issue and not every smoker is an automatic culprit, when it happens, it reveals a generalized lack of care of public spaces that can indeed be expressed in many other ways (*e.g.,* littering of other items). In this case, more than one societal segment must be involved and precautionary actions taken. Legal and educational measures should guarantee behavioural changes in society and industry to enforce policies that involve manufacturers in participating in the costs of pollution control (Smith and Novotny, 2011) and the search for more friendly products and client information (Novotny and Zhao, 1999).

The tobacco industry supports the replacement of the present filter by biodegradable ones, so it would not interrupt or decrease business. However, this would only be a palliative since reducing the residence time in nature does not solve any of the issues discussed in this review. There are no benefits in using biodegradable filters, and it can speed up the transfer of toxic substances to the water and biota as the filter decays, and reduce smoker's guilt when littering. All issues raised in this review (concerning tobacco production, consumption and debris

discard) would remain exactly the same since CB would still be produced and littered in natural environments.

Taxation of cigarettes and other tobacco products are already in the price of the product and seldom discriminated on the price tag or ticket. So, smokers do not notice them, but only changes in total price. Moreover, according to Smith and McDaniel (2011), taxes should be based on the real costs of cleaning and environmental compensation. In the case of an "environmental tax" on cigarettes, this value should be clearly discriminated to buyers, and mechanisms to guarantee the transparency of reports on where this resource was efficiently used should be implemented.

The amount paid as tax could be directed to broader actions such as financing educational campaigns to reduce cigarette butts litter, infrastructure to collect waste or even funding the treatment of diseases related to smoking. Already the amount paid as a fee could be used for more specific proposals such as covering the costs of public cleaning actions.

Chemical addicts of low income can have their income compromised by cigarette consumption. In this case, lower cost illegal cigarettes without tax components are used as an alternative to satiate the addiction. Illegal cigarette markets are a widespread plague all over the world (Boeira, 2002) and illegal products do not have quality control, usually containing more nicotine and tar than usual, to guarantee conservation of tobacco for longer periods. These are usually the social strata with less qualification and more vulnerable heath. So an important component to help decreasing CB in coastal environments, including popular leisure beaches, would be to combat and reduce the trade in illegal tobacco products.

When considering exclusively the environmental aspects of CB littering, observations of the amount of this item on beaches (regardless of their source), we conclude that the situation is unacceptable. Knowing the real dimension of the problem was possible thanks to NGOs clean-up campaigns and massive public engagement in these actions. The reports from these programs bring astonishing numbers and the staggering information that this type of litter is the most frequent of all types of plastic on any beach around the world, considered one of the "big four" (CB, plastic bottles, plastic bottle caps, food wrappers and plastic bags). However, although these coordinated actions have built an enormous databank on beach litter with readily available temporal and spatial information, year after year, the amounts of litter continue to increase, suggesting that the effect of the campaign is timid in terms of changing attitudes of the largest part of the population, remaining closer to a palliative/demonstrative exercise than a real change in societal paradigm. The tide might be starting to turn now, after decades of work, if one observes the mood about beach/marine litter in social media.

Beach and city cleaning are expensive (Schneider et al., 2011; Araújo and Costa, 2006; Ballance et al., 2000) and paid for with public funds that could be used for a number of other important public services (health, education, science, pensions, *etc.)*. Beach and city cleaning are even more critical in developing and underdeveloped countries where tourism may be a source of income to coastal communities because they drain scarce resources from essential services (*e.g.,* food security). In this case, reducing cigarette butts and litter, in general, on the beach by preventing litter from reaching the environment in the first place is the only option.

3.1. Suggestions of study strategies for progress in CB research in coastal environments

The contribution of each of the identified sources (pedestrians, drainage, beach users) to the CB amounts found in coastal environments remains unclear, reflecting the lack of scientifically planned sampling strategies that can detect sources and their changes along time and across coastal spaces (Underwood et al., 2017; Costa et al., 2017). In this way, we suggest that CB research in coastal environments that aim at source identification and subsequent control include statistically sound techniques that allow detection of source positions and intensities. This strategy might generate data that better elucidate the situation highlighted by more general sampling actions based on citizen participation. In addition, further exploitation of the existing data is possible, considering that the large amount of this type of data allows a number of different scientific questions to be posed that can use only part of the most consistent matrices to feed conceptual, statistical and numerical models.

Additionally, this review revealed the need to consider all opportunities to use toxicological tests in more relevant animal models to represent coastal environments of all climates (temperate, tropical) and physical characteristics (estuaries, coastal water, benthos, beaches *etc.)*, which is important for facing the temporal and geographical changes imposed by/to the tobacco industry on the cigarette composition and prevailing target markets.

4. Conclusion

The pollution caused by CB is ubiquitous and environmentally hazardous, and generates economic and social costs.

The aim of this review was to bring together the complexity of the issue to kick start a discussion on how to act in a coordinated way to reduce this type of pollution on beaches and other coastal and aquatic environments. We call for the attention of stakeholders for each part to reduce this grave environmental risk.

Reducing beach/marine pollution from tobacco products is a complex issue that requires interventions beyond environmental stakeholders and conventional comfort zones, but it must be done. We believe the way forward is convincing people of the toxicity of cigarettes and their ability damage health as well as their risks to the environment and regional economies. These problems must be addressed at the same level of importance as lung cancer or mortality, and the tobacco industry must assume its share of pollution prevention/cleaning/compensation tasks. The restrictions to commerce and consumption must continue and, despite ethical individual issues, empathy must be exercised and the environment be regarded as a common good from which all society, including ex-smokers, will benefit.

The solution of the problem requires, beyond a collective effort, more restrictive laws and environmental campaigns focused on smokers. There really should also be further review of a filter ban as an intervention, especially in light of the recent EU Parliament decision to ban single use plastics, including cigarette filters. Finally, reducing CB littering will, without doubt, have an extended benefit on other types of litter as well.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.envres.2019.02.005.

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